

MS&E 342: Stochastic Systems and Learning Theory with Applications in Finance

Lecture 15-17: Continuous-Time Stochastic Games, Mean Field Games, and Mean Field Control

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1 Motivation and module overview

The previous lectures developed continuous-time stochastic control for a single decision-maker. In many financial, economic, and learning problems, however, decisions are made by several strategic agents. For example,

- A trader faces the competition from other traders,
- an insurer designs incentives for an agent whose effort is hidden, or
- a platform coordinates or competes with a large population of users.

This module introduces three related but distinct frameworks:

- **Two-player stochastic games and principal–agent problems:** strategic interaction between a small number of decision-makers, including zero-sum, nonzero-sum, Nash, saddle-point, and Stackelberg/incentive-compatible formulations.
- **Mean field games:** equilibrium models for a large number of individually negligible agents. Each agent optimizes against an aggregate distribution, and the aggregate distribution must be consistent with the optimal behavior.
- **Mean field control:** social-planner or team optimization for a large population. A controller optimizes the evolution of a distribution directly, often through McKean–Vlasov dynamics.

The purpose is not to prove the full theory, but to give students a usable vocabulary: what is being optimized, who takes whose behavior as given, what equilibrium means, and how the main equations differ.

Topic	Main object	Main equilibrium or optimality idea
Two-player game	Controlled diffusion with two controls (α^1, α^2)	Nash equilibrium, saddle point, value of a zero-sum game, or Stackelberg equilibrium
Principal–agent	Contract ξ chosen by principal; hidden action a chosen by agent	Agent incentive compatibility and participation; principal optimizes over admissible contracts
Mean field game	Representative agent plus flow of measures (m_t)	Agent is optimal given m , and $m_t = \mathcal{L}(X_t^*)$
Mean field control	McKean–Vlasov controlled dynamics depending on $\mathcal{L}(X_t)$	A planner optimizes over controls and the induced law flow

Throughout the module, let

$$(\Omega, \mathcal{F}, \mathbb{F} = (\mathcal{F}_t)_{0 \leq t \leq T}, \mathbb{P})$$

be a filtered probability space supporting the Brownian motions needed in each model. The state process is denoted by X , the time horizon by T , and the control sets by A , A_1 , or A_2 . We use m_t , μ_t , or $\mathcal{L}(X_t)$ to denote a probability distribution at time t .

The following notation appears repeatedly:

- α_t or $a(t, X_t)$: a control or feedback action;
- $J_i(\alpha^1, \alpha^2)$: payoff of player i ;

- V , u , or v : value function;
- H : Hamiltonian, whose sign convention depends on whether the problem is a maximization or minimization problem;
- m_t : distribution of a representative agent or population state;
- $\mu_t^N = \frac{1}{N} \sum_{i=1}^N \delta_{X_t^i}$: empirical distribution of an N -player system.

Remark 1.1 (Sign conventions). *Some authors maximize rewards, while others minimize costs. The equations change sign, but the economic content is the same. In these notes, each subsection states its convention explicitly. For teaching, it is often helpful to ask students: “Who chooses what, and is this player maximizing or minimizing?”*

2 Two-player stochastic games and principal–agent problems

This section contains two parts. First, it introduces equilibrium language for continuous-time games with a small number of strategic players. Second, unlike a standard stochastic control problem, the principal does not directly choose the control a_t . The agent chooses a_t , and effort is hidden. The principal can only choose a contract based on observable output, and this contract must make the desired effort incentive compatible.

2.1 Notation, equilibria, and equations

Consider two players choosing controls α^1 and α^2 . The controlled state is

$$dX_t = b(t, X_t, \alpha_t^1, \alpha_t^2) dt + \sigma(t, X_t, \alpha_t^1, \alpha_t^2) dW_t, \quad X_0 = x, \quad (1)$$

and player i receives payoff

$$J_i(t, x; \alpha^1, \alpha^2) = \mathbb{E} \left[\int_t^T f_i(s, X_s, \alpha_s^1, \alpha_s^2) ds + g_i(X_T) \right]. \quad (2)$$

An *open-loop* control is an $\mathbb{F}$ -progressively measurable process. A *Markov feedback* strategy is a map $a_i(t, x)$ with $\alpha_t^i = a_i(t, X_t)$. The distinction matters: feedback strategies can react to the current state, and therefore can react indirectly to previous deviations.

Definition 2.1 (Nash equilibrium). *A pair $(\alpha^{1,*}, \alpha^{2,*})$ is a Nash equilibrium if neither player can improve by deviating unilaterally:*

$$J_1(\alpha^{1,*}, \alpha^{2,*}) \geq J_1(\alpha^1, \alpha^{2,*}) \quad \text{for all } \alpha^1,$$

and

$$J_2(\alpha^{1,*}, \alpha^{2,*}) \geq J_2(\alpha^{1,*}, \alpha^2) \quad \text{for all } \alpha^2.$$

Equivalently, each equilibrium strategy is a best response to the other:

$$\alpha^{1,*} \in \text{BR}_1(\alpha^{2,*}), \quad \alpha^{2,*} \in \text{BR}_2(\alpha^{1,*}).$$

Zero-sum game. In a zero-sum game, player 1 maximizes and player 2 minimizes the same payoff J . The lower and upper values are

$$V^-(t, x) = \sup_{\alpha^1} \inf_{\alpha^2} J(t, x; \alpha^1, \alpha^2), \quad V^+(t, x) = \inf_{\alpha^2} \sup_{\alpha^1} J(t, x; \alpha^1, \alpha^2). \quad (3)$$

Always $V^- \leq V^+$. If equality holds, the game has a value. A stronger object is a *saddle point*: $(\alpha^{1,*}, \alpha^{2,*})$ is a saddle point if

$$J(\alpha^1, \alpha^{2,*}) \leq J(\alpha^{1,*}, \alpha^{2,*}) \leq J(\alpha^{1,*}, \alpha^2)$$

for all admissible deviations. Thus player 1 cannot improve by changing α^1 , and player 2 cannot lower the payoff by changing α^2 .

For a smooth candidate value, define the local Hamiltonian

$$G(t, x, a_1, a_2, p, M) = b(t, x, a_1, a_2) \cdot p + \frac{1}{2} \text{Tr}(\sigma \sigma^\top(t, x, a_1, a_2) M) + f(t, x, a_1, a_2).$$

The lower and upper Hamiltonians are

$$H^-(t, x, p, M) = \sup_{a_1 \in A_1} \inf_{a_2 \in A_2} G(t, x, a_1, a_2, p, M),$$

$$H^+(t, x, p, M) = \inf_{a_2 \in A_2} \sup_{a_1 \in A_1} G(t, x, a_1, a_2, p, M).$$

The *Isaacs condition* is the equality $H^- = H^+$. When it holds, the formal Hamilton–Jacobi–Isaacs equation is

$$\partial_t V(t, x) + H(t, x, \nabla V(t, x), \nabla^2 V(t, x)) = 0, \quad V(T, x) = g(x). \quad (4)$$

Remark 2.2 (What the Isaacs condition means). *The Isaacs condition is a local version of “the order of play does not matter.” If the maximizer first chooses a_1 and the minimizer then chooses a_2 , one obtains H^- . If the minimizer moves first, one obtains H^+ . Equality says the local game has no advantage from moving second.*

Non-zero-sum game. In a nonzero-sum game there is generally no single value function. In a Markovian feedback equilibrium, one looks for controls

$$\alpha_t^{1,*} = a_1^*(t, X_t), \quad \alpha_t^{2,*} = a_2^*(t, X_t),$$

and value functions V_1, V_2 satisfying the coupled HJB system

$$\begin{aligned} 0 &= \partial_t V_1 + \sup_{a_1 \in A_1} \left\{ \mathcal{L}^{a_1, a_2^*} V_1 + f_1(t, x, a_1, a_2^*) \right\}, \\ 0 &= \partial_t V_2 + \sup_{a_2 \in A_2} \left\{ \mathcal{L}^{a_1^*, a_2} V_2 + f_2(t, x, a_1^*, a_2) \right\}, \end{aligned} \quad (5)$$

with $V_i(T, x) = g_i(x)$. The controls are coupled through the best-response conditions, not through a common objective.

Principal–agent problem. A principal–agent problem is also a two-player problem, but the timing and information are different. The principal first offers a contract ξ , usually measurable with respect to observed output. The agent then chooses hidden effort a . In a basic hidden-action model,

$$dX_t = b(a_t) dt + \sigma dW_t, \quad X_0 = x_0, \quad (6)$$

where X is observable output but a_t is not directly observed. A contract is *incentive compatible* for a^* if

$$a^* \in \arg \max_{a \in \mathcal{A}} J_A(a; \xi),$$

and it satisfies *participation* if $J_A(a^*; \xi) \geq R_A$. The principal maximizes over contracts satisfying both constraints. This is a Stackelberg problem: the principal moves first and anticipates the agent's best response.

2.2 Examples

Here we introduce a few examples to demonstrate the concepts introduced earlier.

Example	Key message
Robust log-utility portfolio	A zero-sum HJI can reduce to a local max-min problem; ambiguity acts like extra risk.
Relative-performance LQ game	A nonzero-sum game has coupled best responses; the opponent's feedback changes the Riccati equation.
Linear contract with CARA agent	A contract slope is simultaneously an incentive device and a risk-sharing device.

Example 2.3 (Robust portfolio choice as a zero-sum game). *Let $\lambda = \mu - r$ be the excess return. An investor chooses a portfolio fraction π_t , while an adversarial model chooses a drift distortion θ_t :*

$$\frac{dX_t}{X_t} = (r + \pi_t(\lambda + \theta_t)) dt + \pi_t \sigma dW_t.$$

For log utility, consider the robust objective

$$\sup_{\pi} \inf_{\theta} \mathbb{E} \left[\log X_T + \int_0^T \frac{1}{2\eta} \theta_t^2 dt \right], \quad \eta > 0.$$

The penalty prevents nature from choosing arbitrarily large distortions. Since

$$d \log X_t = \left(r + \pi_t(\lambda + \theta_t) - \frac{1}{2} \sigma^2 \pi_t^2 \right) dt + \pi_t \sigma dW_t,$$

the local max-min problem is

$$\sup_{\pi} \inf_{\theta} \left\{ r + \pi(\lambda + \theta) - \frac{1}{2} \sigma^2 \pi^2 + \frac{1}{2\eta} \theta^2 \right\}.$$

For a fixed π , nature solves

$$\inf_{\theta} \left\{ \pi \theta + \frac{1}{2\eta} \theta^2 \right\}, \quad \theta^*(\pi) = -\eta \pi.$$

Substitution gives the investor's reduced problem

$$\sup_{\pi} \left\{ r + \lambda\pi - \frac{1}{2}(\sigma^2 + \eta)\pi^2 \right\},$$

so

$$\pi^* = \frac{\lambda}{\sigma^2 + \eta}, \quad \theta^* = -\eta \frac{\lambda}{\sigma^2 + \eta}. \quad (7)$$

The value is

$$V(t, x) = \log x + \left(r + \frac{\lambda^2}{2(\sigma^2 + \eta)} \right) (T - t).$$

Remark 2.4. *Connction to Merton's problem. This example is a robust version of Merton's portfolio problem. The investor chooses the risky fraction π_t , but is worried that the true drift may be worse than the estimated drift. The variable θ_t represents this adverse drift distortion.*

For a fixed portfolio π , nature chooses θ to make the investor's objective worse, but large distortions are penalized. In the log-utility case, the local expression involving θ is

$$\pi\theta + \frac{1}{2\eta}\theta^2.$$

Minimizing over θ gives

$$\theta^*(\pi) = -\eta\pi.$$

Thus, if the investor takes a larger risky position, nature responds by pushing the drift more negatively.

Substituting $\theta^*(\pi)$ back into the investor's problem gives the local objective

$$r + \pi(\mu - r) - \frac{1}{2}(\sigma^2 + \eta)\pi^2.$$

Therefore the robust optimal portfolio is

$$\pi^* = \frac{\mu - r}{\sigma^2 + \eta}.$$

Compare this with the classical Merton fraction

$$\pi^{\text{Merton}} = \frac{\mu - r}{\sigma^2}.$$

The robust investor behaves as if volatility were larger. Model uncertainty therefore reduces the risky investment.

Example 2.5 (A relative-performance LQ game). *Two managers control their own positions:*

$$dX_t^i = a_t^i dt + \sigma dW_t^i, \quad i = 1, 2,$$

and each dislikes being far from the other. Let $Y_t = X_t^1 - X_t^2$. Then

$$dY_t = (a_t^1 - a_t^2) dt + \sqrt{2}\sigma dB_t.$$

Player 1 minimizes

$$J_1(a^1, a^2) = \mathbb{E} \left[\int_0^T \left(\frac{1}{2}(a_t^1)^2 + \frac{q}{2}Y_t^2 \right) dt + \frac{g}{2}Y_T^2 \right],$$

and player 2 has the symmetric cost with a^2 in the effort term. A symmetric feedback equilibrium should push the two states together, so try

$$a_1^*(t, y) = -P(t)y, \quad a_2^*(t, y) = P(t)y.$$

If player 2 uses $a_2^*(t, y) = P(t)y$, player 1's HJB equation is

$$0 = \partial_t V_1 + \inf_{a_1} \left\{ (a_1 - P(t)y) \partial_y V_1 + \sigma^2 \partial_{yy} V_1 + \frac{1}{2} a_1^2 + \frac{q}{2} y^2 \right\}.$$

With the quadratic ansatz

$$V_1(t, y) = \frac{1}{2} P(t) y^2 + r(t),$$

the minimizer is

$$a_1^*(t, y) = -\partial_y V_1(t, y) = -P(t)y.$$

Matching coefficients gives

$$P'(t) - 3P(t)^2 + q = 0, \quad P(T) = g, \tag{8}$$

and

$$r'(t) + \sigma^2 P(t) = 0, \quad r(T) = 0.$$

The same calculation for player 2 gives $a_2^*(t, y) = P(t)y$.

Remark 2.6 (Coupling between the two HJBs). *This example is meant to show why a nonzero-sum game is different from a single-agent control problem. Player 1 optimizes while taking player 2's feedback rule as given, and player 2 does the same with player 1's rule. Therefore the equilibrium is obtained by solving two best-response problems simultaneously. The coefficient 3 is not the main point; the main point is the coupling between the two HJB equations.*

Example 2.7 (Linear contract: incentives versus insurance). *The principal observes output X , but does not observe the agent's effort a . Output is modeled by*

$$dX_t = a_t dt + \sigma dW_t, \quad X_0 = x_0,$$

and exerting effort costs the agent

$$c(a) = \frac{\kappa}{2} a^2.$$

The principal offers a linear terminal contract

$$\xi = w_0 + \beta X_T.$$

Here w_0 is a fixed payment and β is the share of output paid to the agent. A larger β gives the agent stronger incentives, but it also makes the agent bear more output risk.

Assume the principal is risk neutral and receives

$$X_T - \xi.$$

Assume the agent has CARA utility with risk aversion $\gamma_A > 0$. For simplicity, restrict attention to constant effort $a_t \equiv a$. Then

$$X_T = x_0 + aT + \sigma W_T.$$

Hence the agent's terminal payment is normally distributed. Under CARA utility, the agent ranks contracts by the certainty equivalent

$$\text{CE}_A(a; w_0, \beta) = w_0 + \beta(x_0 + aT) - \frac{\kappa}{2}a^2T - \frac{\gamma_A}{2}\beta^2\sigma^2T.$$

The first two terms are the expected wage, the third term is the effort cost, and the last term is the risk premium the agent requires for bearing the random part $\beta\sigma W_T$.

For fixed contract parameters (w_0, β) , the agent chooses effort. The terms depending on a are

$$\beta a - \frac{\kappa}{2}a^2.$$

Therefore the agent's best response is

$$a^* = \frac{\beta}{\kappa}.$$

This is the incentive effect of the contract: increasing the slope β induces higher effort.

Let $\bar{R}$ be the agent's reservation certainty equivalent. The agent accepts the contract only if

$$\text{CE}_A(a^*; w_0, \beta) \geq \bar{R}.$$

For a fixed β , the fixed payment w_0 does not affect effort; it only transfers surplus from the principal to the agent. Thus the principal chooses w_0 just large enough to make the agent willing to participate:

$$\text{CE}_A(a^*; w_0, \beta) = \bar{R}.$$

Using $a^* = \beta/\kappa$, this gives

$$w_0^*(\beta) = \bar{R} - \beta x_0 - \frac{\beta^2 T}{2\kappa} + \frac{\gamma_A}{2}\beta^2\sigma^2T.$$

Now the principal only has to choose the incentive slope β . Her expected payoff is

$$\mathbb{E}[X_T - \xi] = \mathbb{E}[(1 - \beta)X_T - w_0].$$

Substituting $a^* = \beta/\kappa$ and $w_0 = w_0^*(\beta)$, we obtain

$$\mathbb{E}[X_T - \xi] = x_0 - \bar{R} + T \left(\frac{\beta}{\kappa} - \frac{1}{2} \left(\frac{1}{\kappa} + \gamma_A \sigma^2 \right) \beta^2 \right).$$

The constant $x_0 - \bar{R}$ does not affect the choice of β . Therefore the optimal incentive slope solves

$$\max_{\beta} \left\{ \frac{\beta}{\kappa} - \frac{1}{2} \left(\frac{1}{\kappa} + \gamma_A \sigma^2 \right) \beta^2 \right\},$$

which gives

$$\beta^* = \frac{1}{1 + \gamma_A \kappa \sigma^2}, \quad a^* = \frac{1}{\kappa(1 + \gamma_A \kappa \sigma^2)}.$$

If effort were observable and contractible, the principal could directly require the agent to choose effort a . Since the principal is risk neutral and the agent can be paid a fixed wage to satisfy participation, the relevant total surplus is

$$x_0 + aT - \frac{\kappa}{2}a^2T - \bar{R}.$$

Thus the principal chooses a to maximize

$$a - \frac{\kappa}{2}a^2,$$

which gives the first-best effort

$$a^{FB} = \frac{1}{\kappa}.$$

With hidden effort, the principal cannot simply tell the agent to choose a^{FB} . The agent chooses effort privately. To motivate effort, the principal must give the agent a share β of output:

$$\xi = w_0 + \beta X_T.$$

The agent's best response is

$$a^* = \frac{\beta}{\kappa}.$$

So implementing the first-best effort would require $\beta = 1$.

However, output is noisy:

$$X_T = x_0 + aT + \sigma W_T.$$

If $\beta > 0$, the agent's wage contains the risky term $\beta\sigma W_T$. Because the agent is risk averse, the principal must compensate the agent for bearing this risk. Therefore a larger β has two effects: it gives stronger incentives, but it also creates a larger risk premium.

The optimal slope is

$$\beta^* = \frac{1}{1 + \gamma_A \kappa \sigma^2}.$$

When $\gamma_A \sigma^2 > 0$, we have $\beta^* < 1$, and hence

$$a^* = \frac{\beta^*}{\kappa} < \frac{1}{\kappa} = a^{FB}.$$

Thus hidden effort creates a second-best outcome: the principal weakens incentives in order to avoid putting too much risk on the agent.

2.3 Principal–agent problems: the incentive-sensitivity principle

The linear contract example suggests that the slope of the contract is the key incentive variable. The following result shows that this is not special to linear contracts. In a continuous-time hidden-action model, the relevant object is the sensitivity of the agent's promised payoff to output.

Consider

$$dX_t = b(a_t) dt + \sigma dW_t, \quad X_0 = x_0,$$

where X is observed but the effort a_t is hidden. The agent is paid a terminal contract ξ , which must be measurable with respect to the output history. For the moment, suppose the agent is risk neutral and evaluates a contract by

$$J_A(a; \xi) = \mathbb{E}^a \left[\xi - \int_0^T c(a_t) dt \right],$$

where $\mathbb{E}^a$ denotes expectation when the agent uses effort a .

Note that a contract is incentive compatible if the recommended effort is the agent's best response.

Proposition 2.8 (Incentive compatibility through output sensitivity). *Fix a recommended effort process a^* . Let Z_t be an output-adapted process and define the contract*

$$\xi = Y_0 + \int_0^T [c(a_t^*) - Z_t b(a_t^*)] dt + \int_0^T Z_t dX_t.$$

Then, for any alternative effort $\tilde{a}$,

$$J_A(\tilde{a}; \xi) = Y_0 + \mathbb{E}^{\tilde{a}} \int_0^T [Z_t b(\tilde{a}_t) - c(\tilde{a}_t) - (Z_t b(a_t^*) - c(a_t^*))] dt.$$

Therefore a^ is incentive compatible if*

$$a_t^* \in \arg \max_{a \in A} \{Z_t b(a) - c(a)\} \quad \text{for a.e. } t.$$

Conversely, if local deviations are allowed, an implemented effort must satisfy this pointwise maximization condition.

Proof. Under an alternative effort $\tilde{a}$,

$$dX_t = b(\tilde{a}_t) dt + \sigma dW_t.$$

Substituting the contract into the agent's payoff gives

$$\begin{aligned} J_A(\tilde{a}; \xi) &= \mathbb{E}^{\tilde{a}} \left[Y_0 + \int_0^T (c(a_t^*) - Z_t b(a_t^*)) dt + \int_0^T Z_t dX_t - \int_0^T c(\tilde{a}_t) dt \right] \\ &= Y_0 + \mathbb{E}^{\tilde{a}} \int_0^T [c(a_t^*) - Z_t b(a_t^*) + Z_t b(\tilde{a}_t) - c(\tilde{a}_t)] dt, \end{aligned}$$

because the stochastic integral has mean zero. This is the displayed formula.

If

$$a_t^* \in \arg \max_{a \in A} \{Z_t b(a) - c(a)\},$$

then the integrand is always less than or equal to zero, so no deviation can give the agent more than Y_0 . Thus a^* is incentive compatible.

For the converse, suppose there is a time and event on which another action a gives a strictly larger value of $Z_t b(a) - c(a)$. Then the agent can deviate only on that event and over a short time interval, which increases the payoff. Hence an implemented effort must satisfy the pointwise condition. $\square$

This proposition gives the main lesson. The principal does not directly choose a_t . Instead, the principal chooses the contract sensitivity Z_t . The agent then chooses the effort that maximizes

$$Z_t b(a) - c(a).$$

Thus Z_t is the dynamic version of the linear-contract slope β .

If b and c are differentiable and the maximizer is interior, the incentive-compatibility condition becomes

$$Z_t b'(a_t^*) = c'(a_t^*).$$

For the quadratic example $b(a) = a$ and $c(a) = \kappa a^2/2$, this gives

$$a_t^* = \frac{Z_t}{\kappa}.$$

So the earlier formula $a^* = \beta/\kappa$ is just the special case $Z_t = \beta$.

Now consider the principal's problem. If the participation constraint is

$$J_A(a^*; \xi) \geq \bar{R},$$

then the contract above gives the agent payoff Y_0 under the recommended effort. Since Y_0 is a transfer from the principal to the agent, the principal chooses

$$Y_0 = \bar{R}.$$

Under the recommended effort, the expected wage is

$$\mathbb{E}^{a^*}[\xi] = Y_0 + \mathbb{E}^{a^*} \int_0^T c(a_t^*) dt.$$

Therefore the principal's expected payoff is

$$x_0 - \bar{R} + \mathbb{E}^{a^*} \int_0^T [b(a_t^*) - c(a_t^*)] dt.$$

Thus, with a risk-neutral agent and no additional constraints, hidden effort alone does not create a distortion. The principal can implement the first-best effort

$$a^{FB} \in \arg \max_{a \in A} \{b(a) - c(a)\}$$

by choosing Z_t so that

$$a^{FB} \in \arg \max_{a \in A} \{Z_t b(a) - c(a)\}.$$

The distortion appears when incentives are costly. For example, if the agent is risk averse, making the contract sensitive to output exposes the agent to Brownian noise. In the CARA-normal case, a contract with output sensitivity Z_t carries the risk premium

$$\frac{\gamma_A}{2} \sigma^2 Z_t^2.$$

The principal's reduced local problem becomes

$$\max_{a, z} \left\{ b(a) - c(a) - \frac{\gamma_A}{2} \sigma^2 z^2 \right\}, \quad a \in \arg \max_{\tilde{a}} \{z b(\tilde{a}) - c(\tilde{a})\}.$$

For an interior solution,

$$z = \frac{c'(a)}{b'(a)}.$$

Hence the principal chooses effort by solving

$$\max_a \left\{ b(a) - c(a) - \frac{\gamma_A}{2} \sigma^2 \left(\frac{c'(a)}{b'(a)} \right)^2 \right\}.$$

This formula is the general incentive-insurance trade-off. The first two terms are total surplus. The last term is the cost of giving the agent enough output sensitivity to implement effort a .

In the quadratic case $b(a) = a$, $c(a) = \kappa a^2/2$, we have $z = \kappa a$. The reduced problem is

$$\max_a \left\{ a - \frac{\kappa}{2} a^2 - \frac{\gamma_A}{2} \sigma^2 \kappa^2 a^2 \right\}.$$

The first-order condition gives

$$1 - \kappa a - \gamma_A \sigma^2 \kappa^2 a = 0,$$

so

$$a^* = \frac{1}{\kappa(1 + \gamma_A \kappa \sigma^2)}.$$

Equivalently, the required output sensitivity is

$$z^* = \kappa a^* = \frac{1}{1 + \gamma_A \kappa \sigma^2}.$$

This recovers the linear-contract formula, but now we see the general principle behind it: higher effort requires higher output sensitivity, and higher output sensitivity is costly when the agent is risk averse and output is noisy.

3 Mean-field games

4 Mean-field controls

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